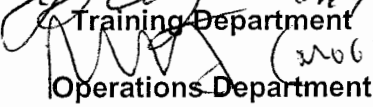


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

STATION:	SALEM		
SYSTEM:	Containment Cooling		
TASK:	Perform a CFCU Operability and Service Water Flow Verification		
TASK NUMBER:	220130201		
JPM NUMBER:	13-01 NRC Sim f		
ALTERNATE PATH:	☐	K/A NUMBER:	022 A4.01
APPLICABILITY:	IMPORTANCE FACTOR:		
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator		
REFERENCES:	S2.OP-ST.CBV-0003 Rev. 17 (Rev checked 9-6-14) S2.OP-SO.CBV-0001 Rev. 32 (Rev checked 9-6-14)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	12 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	9-6-14
Validated By:	D Tait SME or Instructor	Date:	10-15-14
Approved By:	 Training Department	Date:	11-18-14
Approved By:	 Operations Department	Date:	11-18-14
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

NAME: _____

DATE: _____

SYSTEM: Containment Cooling

TASK: Perform a CFCU Operability and Service Water Flow Verification

TASK NUMBER: 220130201

SIMULATOR SETUP: Reset Simulator to IC-256, 100% power, BOL.
ET-1, KAD18PBR 25 CFCU low speed start PB, tied to 25 CFCU SW flow of 1675, ramped in over 10 seconds with 1 second delay

INITIAL CONDITIONS:

Rx power is 100% steady state, BOL.

INITIATING CUE:

You have been directed to perform S2.OP-ST.CBV-0003, Containment Systems-Cooling Systems surveillance for 25 CFCU for monthly run.

25 CFCU is filled and vented.

Component Off Normal Report has been reviewed with no abnormalities noted.

Calibration data for required instruments has been obtained and recorded in appropriate Attachments.

The differential pressure gauge for 22 SW header has been installed.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

Perform S2.OP-ST.CBV-0003 for 25 CFCU and evaluate minimum SW flow as SAT, and Minimum Cooling Water Flow for SW D/P as UNSAT.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: Containment Cooling

TASK: Perform a CFCU Operability and Service Water Flow Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		START TIME: _____			
		Provide Operator with marked up copy of S2.OP-ST.CBV-0003, Containment Systems-Cooling Systems.	Prerequisites have been completed, but Operator should review them before proceeding.		
			Operator reviews and initials Precautions and Limitations.		
			NOTE: There is no requirement to either start additional SW pumps OR stop any running CFCUs. There is a system requirement NOT to run 5 CFCUs in HIGH speed, but that is N/A here. However, Operator MAY start another SW pump OR stop a CFCU, which is acceptable.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: Containment Cooling

TASK: Perform a CFCU Operability and Service Water Flow Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.1.1 SO.CBV -1 5.1.3	PERFORM test on each CFCU required to be tested IAW the following instruction: A. ENSURE the CFCU to be tested is in LOW SPEED IAW S2.OP-SO.CBV-0001, Containment Ventilation Operation. A. ENSURE CFCU is filled and vented IAW S2.OP-SO.CBV-0003(Q), Filling and Venting Containment Fan Coil Units. B. ENSURE Service Water is available. C. PRESS FAN LOW SPEED START bezel. D. ENSURE following dampers are in indicated positions: • ROUGH FLTR DAMPER CLOSED • HEPA INLET DAMPER OPEN • HEPA OUTLET DAMPER OPEN	Operator retrieves S2.OP-SO.CBV-0001, Containment Ventilation Operation, and reviews P&Ls, and Prerequisites. Operator identifies the sections of the procedure NOT to be performed (Sections 5.2-5.6) with N/A. If needed, tell Operator Components Off Normal review is complete and SAT. Operator starts 25 CFCU in LOW speed IAW Step 5.1.3 by: A. Ensuring 25 CFCU is filled and vented (Initiating Cue) B. Ensuring SW is available by checking SW pumps in service and adequate SW header pressure. C. PRESS FAN LOW SPEED START bezel for 25 CFCU. D. ENSURES damper alignment is correct for 25 CFCU: ROUGH FLTR DAMPER CLOSED HEPA INLET DAMPER OPEN HEPA OUTLET DAMPER OPEN		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Containment Cooling

TASK: Perform a CFCU Operability and Service Water Flow Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	SO.CBV -1 5.1.3 (cont)	E. <u>IF</u> Service Water flow is <1465 gpm, <u>THEN</u> : 1. STOP the CFCU. 2. REFER to S2.OP-SO.SW-0005, Service Water System Operation, Service Water System Operability Guidelines. F. ENSURE Service Water flow $\geq$ 1465 gpm.	Verifies SW flow >1465 gpm to 25 CFCU. Note: <u>IF</u> operator notices SW flow of 1675 is not like the running CFCU's and states they would stop the performance of test, <u>THEN Cue</u> that the CRS directs you to continue the test.		
	ST.CBV- 3 5.1.1	B. RECORD Start Time in the applicable Attachments(s), Section 3.0, for the CFCU being tested.	Operator records start time of 25 CFCU in Attachment 5.		

NAME: _____
DATE: _____

TASK: Perform a CFCU Operability and Service Water Flow Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.1.1 (cont)	C. When at least 15 minutes have elapsed, RECORD the following in the applicable Attachment(s), Section 3.0, for the CFCU being tested: • Stop Time • Cooling Water Flow Rate (gpm) • IF testing 21 <u>OR</u> 22 CFCU, <u>THEN</u> 21 SW HDR ΔP is to be recorded. • IF testing 23 CFCU, <u>THEN</u> both 21 SW HDR ΔP <u>AND</u> 22 SW HDR ΔP are to be recorded (Refer to Step 3.5) • IF testing 24 <u>OR</u> 25 CFCU, <u>THEN</u> 22 SW HDR ΔP is to be recorded. • Test results by initialing the SAT or UNSAT column IAW the Acceptance Criteria stated in the attachment.	Cue: 25 CFCU has been operating for 15 minutes. Operator records 25 CFCU stop time, SW flow rate, and 22 SW header ΔP. Operator marks steps for 21-23 CFCU N/A. Cue: WHEN asked, report 22 SW header ΔP is 91 psid. Using Exhibit 1, determines minimum CFCU flowrate for 25 CFCU at 91 psid (required to be rounded up to 95 psid IAW P&L 3.4) Operator initials >1465 gpm as SAT, and Cooling Water Flow Rate as <u>UNSAT</u>. Terminate the JPM once the Att. 5 data in Section 3.0 is marked SAT/UNSAT.		

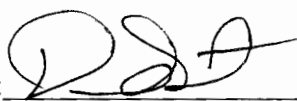
JOB PERFORMANCE MEASURE

TQ-AA-106-0303
Revision 4

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ☒ 1. Task description and number, JPM description and number are identified.
- ☒ 2. Knowledge and Abilities (K/A) references are included.
- ☒ 3. Performance location specified. (in-plant, control room, or simulator)
- ☒ 4. Initial setup conditions are identified.
- ☒ 5. Initiating and terminating Cues are properly identified.
- ☒ 6. Task standards identified and verified by SME review.
- ☒ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ☒ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 17 Date 10/15/14
32
- ☒ 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- ☐ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- ☐ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: 

Date: 10/15/14

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

JOB PERFORMANCE MEASURE

INITIAL CONDITIONS:

Rx power is 100% steady state, BOL.

INITIATING CUE:

You have been directed to perform S2.OP-ST.CBV-0003, Containment Systems-Cooling Systems surveillance for **25** CFCU for monthly run.

25 CFCU is filled and vented.

Component Off Normal Report has been reviewed with no abnormalities noted.

Calibration data for required instruments has been obtained and recorded in appropriate Attachments.

The differential pressure gauge for 22 SW header has been installed.